

Notes: Fajgelbaum, Goldberg, Kennedy & Khandelwal (QJE, 2020)

The Return to Protectionism

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1 Big picture

- **Question.** What were the short-run economic, distributional, and political-economy consequences of the 2018 U.S. tariff increases and foreign retaliations?
- **Empirical setting.** In 2018, the United States imposed large tariff increases across several waves, and major trading partners retaliated against U.S. exports. This created sharp variation across products, source countries, destination markets, and time.
- **Core incidence result.** U.S. tariffs were passed through almost completely into duty-inclusive U.S. import prices. Before-duty foreign export prices did not fall enough to absorb the tariff.
- **Trade quantity result.** Targeted imports and exports fell substantially after tariff increases. The intensive margin response is large even in the short run.
- **Consumer and firm buyer losses.** Higher duty-inclusive import prices generated a large cost to U.S. consumers and firms that buy imported goods, estimated at roughly \$51 billion, or 0.27 percent of GDP.
- **General-equilibrium result.** Once tariff revenue and producer gains are included, the estimated aggregate real income loss is much smaller: around \$7.2 billion, or 0.04 percent of GDP.
- **Redistribution result.** The small aggregate loss masks large redistribution from import buyers to protected producers and the government, as well as large redistribution across regions.
- **Political-economy result.** U.S. tariffs protected industries concentrated in electorally competitive counties, while retaliatory tariffs hit export sectors concentrated in heavily Republican counties.
- **Regional result.** Model simulations imply that all counties lost in tradeable real wage terms, but the largest losses fell on heavily Republican counties because of retaliation.
- **Takeaway.** The trade war generated small aggregate effects but large incidence and distributional effects; the key policy lesson is that who pays and who is protected matters as much as the aggregate welfare number.

2 Incremental contribution of the paper

- **A modern tariff-incidence measurement exercise.** The paper provides one of the first comprehensive product-level empirical estimates of the short-run effects of the 2018 U.S. trade war using the actual timing and product-country coverage of tariff waves.
- **Connecting incidence to welfare.** Rather than stopping at reduced-form trade responses, the paper embeds estimated elasticities into a quantitative general-equilibrium model to translate price and quantity responses into consumer losses, producer gains, tariff revenue, and aggregate equivalent variation.
- **Joint analysis of import tariffs and retaliation.** The paper studies both sides of the trade war: U.S. import protection and foreign retaliation against U.S. exports. This is crucial because the political and regional conclusions look very different once retaliatory tariffs are included.
- **Variety-product-sector nesting.** The empirical framework distinguishes substitution across imported varieties, across imported products, and between imports and domestic output within sectors. This layered structure makes the welfare and pass-through calculations more credible than a single aggregate elasticity.
- **Regional and political incidence.** The paper goes beyond national welfare by mapping product-level tariff exposure into U.S. counties through the local industrial composition of employment and wages.
- **Policy design versus policy incidence.** The paper separates the apparent ex ante political targeting of U.S. tariffs toward competitive counties from the ex post distributional consequences of retaliation, which hurt Republican-leaning counties most.
- **A benchmark for later trade-war research.** Subsequent work on tariffs, pass-through, uncertainty, and the political economy of trade policy uses this paper as a baseline because it combines clean tariff variation, high-frequency trade data, and model-based incidence accounting.

3 Data and measurement

3.1 Tariff waves and product coverage

The paper constructs detailed tariff data for each 2018 U.S. tariff wave and for foreign retaliatory measures against U.S. exports. A trade “variety” is a country-product pair. Products are measured at the HS-10 level for U.S. imports and at closely related product levels for exports.

- U.S. import tariff waves include solar panels, washing machines, aluminum, iron and steel, and several China-specific tariff lists.
- Retaliatory tariffs were imposed by China, Mexico, Turkey, the European Union, Canada, and Russia.
- U.S. tariff increases covered more than 12,000 import products and roughly \$303 billion in annual imports.
- Retaliatory tariffs covered more than 8,000 export products and roughly \$127 billion in annual exports.

Interpretation: The table-level distinction between U.S. tariffs and retaliatory tariffs is essential. Import tariffs affect U.S. consumers and firms that use imported inputs; retaliatory tariffs affect U.S. exporters and their workers.

3.2 Trade data and unit values

The main trade data are monthly U.S. Census trade flows from January 2017 through April 2019. The authors observe values, quantities, and unit values at high frequency and high product-country detail.

The key outcomes are:

$$\Delta \ln(p_{igt}^* m_{igt}), \quad \Delta \ln m_{igt}, \quad \Delta \ln p_{igt}^*, \quad \Delta \ln p_{igt},$$

where i indexes countries, g indexes products, t indexes time, m is quantity, p^* is the before-duty unit value, and p is the duty-inclusive unit value.

Interpretation: The difference between before-duty and duty-inclusive prices identifies incidence. If before-duty prices fall after tariffs, foreign exporters absorb part of the tariff. If duty-inclusive prices rise one-for-one while before-duty prices do not fall, U.S. buyers bear the tariff burden.

3.3 Tariff measures

For each variety, the paper measures changes in statutory tariffs:

$$\Delta \ln(1 + \tau_{igt}).$$

For import regressions, τ_{igt} is the U.S. tariff on imports. For export regressions, τ_{igt}^* is the retaliatory tariff faced by U.S. exports.

Interpretation: Using log changes in one plus the tariff rate makes the coefficients interpretable as elasticities with respect to gross tariff-inclusive costs.

3.4 Product, sector, and county aggregation

The paper aggregates variety-level shocks to higher levels in several steps.

- Product-level import shocks are aggregated to HS products using prewar import shares.

- Sector-level shocks are mapped into NAICS industries.
- County-level tariff exposure is built by weighting industry tariff shocks by each county's 2016 tradeable-sector wage bill.
- Voting data come from 2016 presidential election results and are mapped to counties.

Interpretation: The county exposure measure is not a direct tariff on local consumers. It is an employment-weighted measure of how much local tradeable workers are exposed through the products and sectors in which their county specializes.

4 Models and empirical specifications

4.1 Event-study design

For each tariff wave, the paper estimates dynamic effects around the month of tariff implementation:

$$\Delta \ln y_{igt} = \alpha_{gt} + \alpha_{it} + \alpha_{is} + \sum_{m=-6}^{6+} \beta_m [\ln(1 + \tau_{ig,t-m}) - \ln(1 + \tau_{ig,t-1-m})] + \varepsilon_{igt}.$$

Interpretation: The event study tests whether targeted and untargeted varieties were already on different trends before tariffs and whether trade values, quantities, and prices change after the tariff wave.

4.2 Import variety-level regressions

The core import specification regresses first differences in trade outcomes on tariff changes:

$$\Delta \ln y_{igt} = \alpha_{gt} + \alpha_{it} + \alpha_{is} + \beta \Delta \ln(1 + \tau_{igt}) + \varepsilon_{igt}.$$

- α_{gt} are product-time fixed effects.
- α_{it} are country-time fixed effects.
- α_{is} are country-sector fixed effects.
- y_{igt} is import value, quantity, before-duty price, or duty-inclusive price.

Interpretation: Product-time effects absorb shocks common to all source countries selling a product. Country-time effects absorb exporter-country shocks such as exchange rates. Country-sector effects absorb persistent supply differences across exporters and sectors.

4.3 Product and sector elasticities

The paper estimates the product elasticity η and sector elasticity κ from nested demand relationships. At the product level, the key idea is that tariffs reallocate expenditure across imported products inside sectors. At the sector level, tariffs reallocate expenditure between imported and domestic goods.

$$\begin{aligned} \Delta \ln s_{Mgt} &= \alpha_{st} + (1 - \eta) \Delta \ln P_{Mgt} + u_{gt}, \\ \Delta \ln S_{Mst} &= \alpha_s + \alpha_t + (1 - \kappa) \Delta \ln(P_{Mst}/P_{Dst}) + v_{st}. \end{aligned}$$

Interpretation: The product and sector elasticities determine how much tariff-induced price increases translate into reduced imports and substitution toward untargeted products or domestic output.

4.4 Export regressions for retaliatory tariffs

For U.S. exports hit by foreign retaliation, the analogous specification is:

$$\Delta \ln y_{igt}^X = \alpha_{gt} + \alpha_{it} + \alpha_{is} + \beta \Delta \ln(1 + \tau_{igt}^*) + \varepsilon_{igt}^X.$$

Interpretation: This regression measures how retaliatory tariffs affected U.S. export quantities, prices, and values. It is the export-side mirror image of the import pass-through analysis.

4.5 Welfare decomposition

The paper summarizes aggregate effects using equivalent variation:

$$EV = -\mathbf{m}'\Delta\mathbf{p}^M + \mathbf{x}'\Delta\mathbf{p}^X + \Delta R.$$

- $-\mathbf{m}'\Delta\mathbf{p}^M$ is the loss from higher duty-inclusive import prices.
- $\mathbf{x}'\Delta\mathbf{p}^X$ is the gain to U.S. producers from export and domestic price changes.
- ΔR is tariff revenue.

Interpretation: The large consumer loss does not equal the net national loss because tariff revenue and producer gains partly offset it. The formula also makes clear that tariffs are redistributive taxes plus terms-of-trade and supply-side distortions.

5 Identification and empirical design

5.1 Source of identifying variation

The identifying variation comes from the timing and magnitude of 2018 tariff changes across products, countries, and months. Some products and source countries were hit by large tariff changes; others were not. Some waves happened early in 2018; others happened later.

Interpretation: The empirical design is close to a difference-in-differences/event-study design at the variety-month level: targeted products are compared to untargeted products before and after tariff implementation, with high-dimensional fixed effects absorbing common shocks.

5.2 Why product-time and country-time fixed effects matter

Tariffs were not randomly assigned. Products targeted by the U.S. government may differ from other products. The paper therefore controls for product-time and country-time effects.

- Product-time fixed effects absorb time-varying shocks common to all exporters of a product.
- Country-time fixed effects absorb exporter-country shocks, including exchange-rate and macro shocks.
- Country-sector fixed effects absorb persistent comparative advantage and sector-level sourcing differences.

Interpretation: The coefficient is identified from within-product, within-time, across-country variation and from the differential timing of tariff exposure, rather than from a simple comparison of targeted products to all untargeted products.

5.3 Pretrend tests

The pretrend specification tests whether future tariff changes predict pre-2018 changes in import or export outcomes:

$$\Delta y_{2017} = \alpha + \rho \Delta \ln(1 + \tau_{2018}) + FE + \varepsilon.$$

Interpretation: If tariff-targeted varieties were already declining before the trade war, post-tariff declines would not be credible evidence of tariff effects. Table III shows little evidence of such pretrends.

5.4 Identification of pass-through

Pass-through is identified by comparing before-duty unit values and duty-inclusive unit values after tariff changes.

- If foreign exporters absorb the tariff, before-duty unit values should fall.
- If U.S. buyers absorb the tariff, duty-inclusive unit values should rise while before-duty unit values remain flat.
- The paper finds the second pattern.

Interpretation: This is the paper’s core incidence design. It directly asks who pays the tariff in the short run: foreign exporters or U.S. consumers and importing firms.

5.5 Model-based regional design

The regional analysis is model-based rather than a direct county wage regression. The model maps tariff shocks into county-level changes using each county’s industrial employment structure and the general-equilibrium response of sector prices and wages.

$$\hat{w}_r^T - \hat{P}_r$$

measures the change in county r ’s tradeable real wage.

Interpretation: The model is necessary because a tariff affects counties not only through direct protection or retaliation, but also through input-output linkages, sectoral wage changes, and the local cost of living.

5.6 Threats and limitations

- The analysis is short-run; long-run relocation of sourcing and production may differ.
- The paper does not directly observe retail prices paid by final consumers.
- Tariff uncertainty and investment responses are outside the core framework.
- The model abstracts from imperfect competition and some foreign wage responses.
- Product targeting may reflect political or economic selection, but high-dimensional fixed effects and pretrend tests reduce this concern for short-run trade responses.

Interpretation: The strongest claims are about short-run trade quantities, import-price incidence, and model-implied distributional effects. The paper is more cautious about long-run welfare, investment, and uncertainty channels.

6 Tables and figures

6.1 Figure I: Trade War Timeline; Table I: The 2018 Trade War

Read:

- Figure I shows that tariffs arrived in discrete waves during 2018 rather than as a single policy event.
- Panel A tracks U.S. tariffs on targeted imports; Panel B tracks retaliatory tariffs on U.S. exports.
- Table I lists the product counts, trade values, and tariff rates for each wave.
- The U.S. tariff waves raised average targeted import tariffs to 16.6 percent; retaliation raised targeted export tariffs to 20.4 percent.

Economic message: The timeline and table define the treatment. They show that the trade war is broad enough to matter macroeconomically but staggered enough to provide empirical variation.

Detailed interpretation: The most important reading point is the scale: the policy does not only target symbolic sectors. It covers hundreds of billions of dollars of trade and many thousands of products. The variation across waves helps identify event-study effects, while the final total rows provide the magnitude used in welfare calculations.

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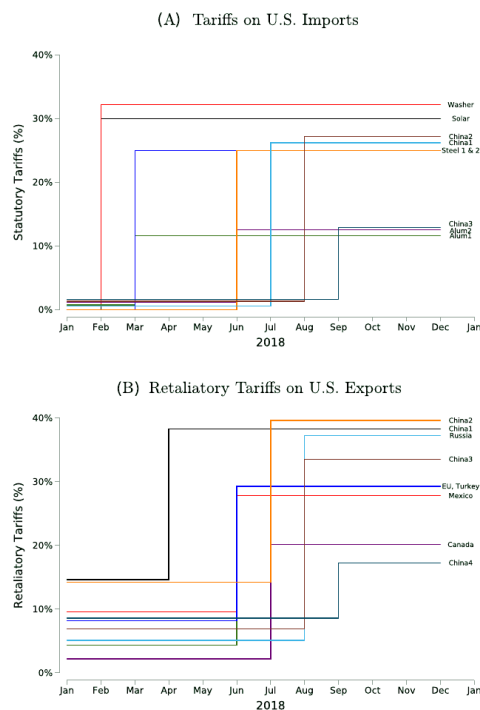


FIGURE I

Trade War Timeline

Figure shows the unweighted average tariff rate of targeted import and export

(a) **Figure I: Trade War Timeline.** Unweighted average tariff rates for targeted imports and exports across 2018 tariff waves.

TABLE I
THE 2018 TRADE WAR

Tariff wave	Date enacted	Products	2017 imports		Tariff (%)	
		(# HS-10)	(mil US\$)	(%)	2017	2018
Panel A: Tariffs on U.S. imports enacted by the United States in 2018						
Solar panels	Feb 7, 2018	8	5,782	0.2	0.0	30.0
Washing machines	Feb 7, 2018	8	2,105	0.1	1.3	32.2
Aluminum	Mar–Jun, 2018	67	17,685	0.7	2.0	12.0
Iron and steel	Mar–Jun, 2018	753	30,523	1.3	0.0	25.0
China 1	Jul 6, 2018	1,672	33,510	1.4	1.3	26.2
China 2	Aug 23, 2018	433	14,101	0.6	2.7	27.0
China 3	Sep 24, 2018	9,102	199,264	8.3	3.3	12.9
Total		12,043	302,970	12.7	2.6	16.6
Panel B: Retaliatory tariffs on U.S. exports enacted by trading partners in 2018						
China	Apr–Sep, 2018	7,474	92,518	6.0	8.4	18.9
Mexico	Jun 5, 2018	232	6,746	0.4	9.6	28.0
Turkey	Jun 21, 2018	244	1,554	0.1	9.7	31.8
European Union	Jun 22, 2018	303	8,244	0.5	3.9	29.2
Canada	Jul 1, 2018	325	17,818	1.2	2.1	20.2
Russia	Aug 6, 2018	163	268	0.0	5.2	36.8
Total		8,073	127,149	8.2	7.3	20.4

Notes. Panels display unweighted monthly HS-10 country average statutory tariff rates. 2017 tariff rates are computed as the annual average; 2018 tariff rates are computed using data from December 2018. Total tariff rates are computed as the trade-weighted average of table row values. The denominator for import (export) share is the total 2017 annual US\$ value of all U.S. imports (exports). The U.S. government announced import tariffs on aluminum and steel products on March 23 but granted exemptions for Canada, Mexico, and the European Union; those exemptions were lifted on June 1. The dates of Chinese retaliations are April 6, July 2, August 23, and September 24. See the text for data sources.

(b) **Table I: The 2018 Trade War.** Product coverage, trade values, and before/after tariff rates for U.S. tariffs and foreign retaliation.

6.2 Table II: Sector Variation in Tariff Rate Changes for Targeted Import Varieties and Export Products by NAICS-3 Code

Read:

- Table II shows that tariff exposure varies substantially across sectors.
- Import tariffs are concentrated in sectors such as machinery, electronics, metal products, and manufactured goods.
- Retaliatory tariffs are heavily present in agricultural and food-related sectors as well as selected manufacturing sectors.
- This sectoral heterogeneity later drives regional heterogeneity because counties specialize in different industries.

Economic message: The table is the bridge from product-level policy to sectoral and regional incidence. If every sector had identical tariff exposure, regional redistribution would be much weaker.

Detailed interpretation: The table also helps explain why retaliation changes the political incidence of the war. U.S. tariffs are concentrated in manufacturing and industrial regions; retaliation hits exports that are more rural and agricultural. This asymmetry becomes central in Figures V–VIII.

Sector (1)	Imports (U.S. tariffs)						Exports (retaliatory tariffs)					
	NAICS-3 (2)	# Products (3)	# Varieties (4)	Δ Tariffs		# Products (7)	# Varieties (8)	Δ Tariffs				
				Mean (5)	Std. dev. (6)			Mean (9)	Std. dev. (10)			
rop and animal production	111-2	456	456	0.10	0.00	303	380	0.24	0.11			
restry and logging	113	71	71	0.10	0.00	79	79	0.12	0.07			
ishing, hunting, and trapping	114	486	486	0.10	0.00	247	247	0.24	0.03			
il and gas extraction	211	17	17	0.10	0.00	8	8	0.22	0.07			
lining (except oil and gas)	212	103	103	0.10	0.00	89	92	0.10	0.05			
ood	311	732	732	0.10	0.00	622	1,014	0.17	0.09			
verage and tobacco products	312	64	64	0.10	0.00	55	379	0.23	0.06			
extile mills	313	1,502	1,502	0.10	0.00	468	494	0.12	0.06			
extile product mills	314	176	176	0.10	0.00	122	235	0.16	0.08			
pparel	315	92	92	0.10	0.00	325	1,082	0.20	0.07			
ather and allied products	316	237	237	0.10	0.00	196	357	0.16	0.08			
ood products	321	424	424	0.10	0.00	194	194	0.10	0.03			
aper	322	335	335	0.12	0.05	239	388	0.12	0.07			
rinting and related activities	323	14	14	0.10	0.00	46	74	0.13	0.09			

(a) **Table II, first page.** Sector exposure for targeted import varieties and export products.

(CONTINUED)									
Sector (1)	Imports (U.S. tariffs)					Exports (retaliatory tariffs)			
	NAICS-3 (2)	# Products (3)	# Varieties (4)	Δ Tariffs		# Products (7)	# Varieties (8)	Δ Tariffs	
				Mean (5)	Std. dev. (6)			Mean (9)	Std. dev. (10)
petroleum and coal products	324	74	74	0.13	0.06	64	64	0.23	0.05
chemical	325	1,730	2,730	0.12	0.05	1,159	1,411	0.12	0.08
plastics and rubber products	326	251	251	0.15	0.07	171	196	0.10	0.07
nonmetallic mineral products	327	354	354	0.11	0.03	225	632	0.18	0.08
primary metal	331	1,147	14,093	0.19	0.07	495	1,738	0.20	0.07
abricated metal products	332	583	852	0.14	0.06	404	1,236	0.18	0.09
machinery	333	1,344	1,344	0.20	0.07	1,075	1,218	0.11	0.06
computer and electronic products	334	617	878	0.21	0.07	458	506	0.11	0.07
electrical equipment and appliances	335	414	594	0.18	0.08	326	656	0.16	0.08
transportation equipment	336	429	429	0.15	0.07	273	680	0.21	0.08
furniture and related products	337	160	160	0.10	0.01	37	244	0.21	0.07
miscellaneous	339	231	231	0.13	0.06	393	608	0.16	0.09
total		12,043	25,699	0.12	0.03	8,073	14,212	0.17	0.07

Notes: Table shows the mean and standard deviation of tariff increases across 3-digit NAICS sectors. A tariff change of 0.10 indicates a 10 percentage point increase. Sectors with a small number of targeted varieties and products in columns (3) and (4) reflect import tariffs exclusively targeting Chinese products. Means and standard deviations in the final row are computed as the simple average of table row values. Import tariffs are constructed from U.S. International Trade Commission (USITC) documents, and retaliatory tariffs are constructed using official documents from foreign finance and trade ministries.

(b) **Table II, continued.** Remaining sectors and total rows for tariff exposure.

6.3 Figure II: Variety Event Study for Imports; Figure III: Variety Event Study for Exports

Read:

- Figure II shows sharp post-tariff declines in targeted import values and quantities.
- Figure II also shows little decline in before-duty import unit values and a clear increase in duty-inclusive unit values.
- Figure III shows that retaliatory tariffs reduce U.S. export values and quantities.
- The pre-period coefficients are relatively flat, supporting the event-study identification.

Economic message: These event studies provide the visual evidence for the main incidence result: tariffs reduce quantities, and U.S. import tariffs are passed through into buyer prices rather than absorbed by foreign exporters.

Detailed interpretation: The difference between the bottom-left and bottom-right panels of Figure II is the cleanest visual evidence. Before-duty unit values barely move, but duty-inclusive values rise after tariffs. This means the tariff wedge is passed to U.S. buyers in the short run.

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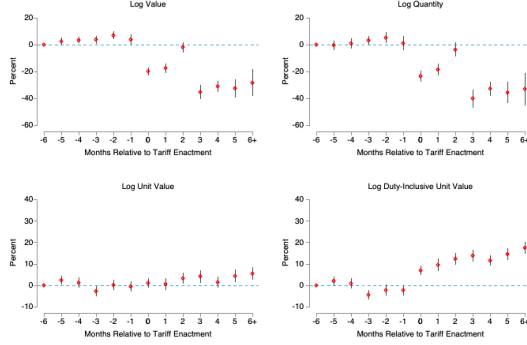


FIGURE II
Variety Event Study: Imports

Figure plots event time dummies for targeted varieties relative to untargeted varieties. Regressions include country-product, product-time, and country-time fixed effects. Standard errors are clustered by country and HS-8. Event periods before -6 are dropped, and event periods ≥ 6 are binned. Error bars show 95% confidence intervals. In [Online Appendix B](#) we provide evidence that the temporary surge in imports during event period 2 reflects an anticipation response to additional tariff threats on a subset of Chinese varieties. Sample: monthly variety-level import data are from U.S. Census. The sample period is 2017:1 to 2019:4.

reveals anticipatory effects occurring before the tariff changes.

(a) **Figure II: Variety Event Study for Imports.** Dynamic effects of U.S. import tariffs on import values, quantities, and unit values.

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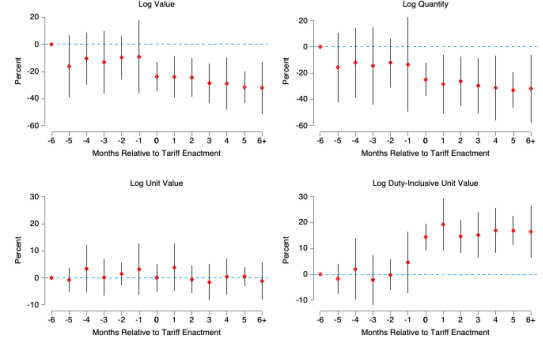


FIGURE III
Variety Event Study: Exports

Figure plots event time dummies for targeted varieties relative to untargeted varieties. Regressions include country-product, product-time, and country-time fixed effects. Standard errors are clustered by country and HS-6. Event periods before -6 are dropped, and event periods ≥ 6 are binned. Error bars show 95% confidence intervals. Sample: monthly variety-level export data are from U.S. Census. The sample period is 2017:1 to 2019:4.

III. TRADE FRAMEWORK AND IDENTIFICATION

Here we describe the trade framework that guides the estimation. We defer supply-side and general-equilibrium assumptions

(b) **Figure III: Variety Event Study for Exports.** Dynamic effects of retaliatory tariffs on U.S. export values, quantities, and unit values.

6.4 Table III: Tests for Preexisting Trends; Table IV: Variety Import Demand and Foreign Export Supply

Read:

- Table III shows no statistically meaningful prewar trends in targeted import or export outcomes.
- Table IV shows import values and quantities fall almost one-for-one with tariff-inclusive costs.
- Table IV shows no significant fall in before-duty import unit values.
- Duty-inclusive import prices rise, implying short-run tariff pass-through to U.S. buyers.

Economic message: Table III protects the design from the concern that tariffed varieties were already trending differently. Table IV quantifies the central incidence and elasticity estimates.

Detailed interpretation: Table IV is the key empirical table. The coefficient near zero on before-duty unit values says foreign exporters did not cut prices to offset tariffs. The positive coefficient on duty-inclusive prices says buyers paid more. The negative quantity coefficient gives the demand response that feeds into welfare and elasticity estimates.

TABLE III
TESTS FOR PREEXISTING TRENDS

	$\Delta \ln p_{ig}^* m_{ig}$ (1)	$\Delta \ln m_{ig}$ (2)	$\Delta \ln p_{ig}^*$ (3)	$\Delta \ln p_{ig}$ (4)
Panel A: U.S. import trends				
$\Delta_{17-18} \ln(1 + \tau_{ig})$	0.12 (0.11)	-0.04 (0.19)	0.18 (0.15)	0.18 (0.15)
Country $\times$ sector FE	Yes	Yes	Yes	Yes
Product FE	Yes	Yes	Yes	Yes
R^2	0.14	0.14	0.14	0.14
N	180,744	149,173	149,173	149,173
	$\Delta \ln p_{ig}^X x_{ig}$	$\Delta \ln x_{ig}$	$\Delta \ln p_{ig}^X$	$\Delta \ln p_{ig}^X (1 + \tau_{ig}^*)$
Panel B: U.S. export trends				
$\Delta_{17-18} \ln(1 + \tau_{ig}^*)$	0.07 (0.06)	0.11 (0.09)	-0.03 (0.07)	-0.03 (0.07)
Country $\times$ sector FE	Yes	Yes	Yes	Yes
Product FE	Yes	Yes	Yes	Yes
R^2	0.11	0.12	0.12	0.12
N	207,840	163,181	163,181	163,181

Notes. Table reports pretrend tests for import (Panel A) and export (Panel B) variety-level trade outcomes. Table reports regressions of the 2017:1–2017:12 average monthly changes in values, quantities, unit values, and tariff-inclusive unit values against the 2018 tariff changes. Standard errors are clustered by country and

(a) **Table III: Tests for Preexisting Trends.** Prewar import and export trends are not systematically related to future tariffs.

TABLE IV
VARIETY IMPORT DEMAND (σ) AND FOREIGN EXPORT SUPPLY (ω^*)

	$\Delta \ln p_{igt}^* m_{igt}$ (1)	$\Delta \ln m_{igt}$ (2)	$\Delta \ln p_{igt}^*$ (3)	$\Delta \ln p_{igt}$ (4)	$\Delta \ln p_{igt}^*$ (5)	$\Delta \ln m_{igt}$ (6)
$\Delta \ln(1 + \tau_{igt})$	-1.52*** (0.18)	-1.47*** (0.24)	0.00 (0.08)	0.58*** (0.13)		
$\Delta \ln m_{igt}$					-0.00 (0.05)	
$\Delta \ln p_{igt}$						-2.53*** (0.26)
Product $\times$ time FE	Yes	Yes	Yes	Yes	Yes	Yes
Country $\times$ time FE	Yes	Yes	Yes	Yes	Yes	Yes
Country $\times$ sector FE	Yes	Yes	Yes	Yes	Yes	Yes
1st-stage F					36.5	21.2
Bootstrap CI					[-0.14, 0.10]	[1.75, 3.02]
R^2	0.13	0.13	0.11	0.11	0.00	—
N	2,993,288	2,454,023	2,454,023	2,454,023	2,454,023	2,454,023

Notes. Table reports the variety-level import responses to import tariffs. Columns (1)–(4) report import values, quantities, before-duty unit values, and duty-inclusive unit values regressed on the statutory tariff rate. Column (5) reports the foreign export supply curve IV regression, $\hat{\phi}^*$, from equation (9); the first stage is column (2). Column (6) reports the import demand curve IV regression, $\hat{\sigma}$, from equation (8); the first stage is column (4). All regressions include product-time, country-time, and country-sector fixed effects. The coefficient in column (4) is not 1 plus the coefficient in column (3) because the duty inclusive unit value is constructed using actual duties collected by U.S. customs data. Standard errors are clustered by country and

(b) **Table IV: Variety Import Demand and Foreign Export Supply.** Tariff effects on import values, quantities, before-duty prices, and duty-inclusive prices.

6.5 Table V: Product Elasticity; Table VI: Sector Elasticity

Read:

- Table V estimates the product-level elasticity η using product import shares and price indexes.
- The preferred product elasticity is about 1.53.
- Table VI estimates the sector-level elasticity κ , around 1.19.
- Both estimates are used in the nested demand and welfare model.

Economic message: The elasticities determine how strongly buyers substitute away from tariffed goods. These parameters control the size of quantity responses, price indexes, and welfare losses.

Detailed interpretation: The product elasticity governs substitution across imported products within a sector; the sector elasticity governs substitution between imported and domestic goods. The estimates being modest means tariff hikes reduce imports but do not produce complete short-run rerouting or domestic substitution.

TABLE V
PRODUCT ELASTICITY η

	$\Delta \ln s_{Mgt}$ (1)	$\Delta \ln p_{Mgt}$ (2)	$\Delta \ln s_{Mgt}$ (3)
$\Delta \ln Z_{Mgt}$	-0.81** (0.39)	1.52*** (0.40)	
$\Delta \ln p_{Mgt}$			-0.53* (0.27)
Sector-time FE	Yes	Yes	Yes
1st-stage F			14.6
$\hat{\eta}$ (se[$\hat{\eta}$])			1.53 (0.27)
Bootstrap CI			[1.15, 1.89]
R^2	0.01	0.10	—
N	371,916	371,916	371,916

Notes. Table reports the product-level import responses to import tariffs. Column (1) reports the reduced-form regression of the imported product's share within sectoral imports, s_{gt} , on the product-level instrument, Z_{gt} . Column (2) reports the first stage: the regression of the product-level import price index P_{gt} on Z_{gt} . Column (3) reports the IV regression with the implied $\hat{\eta}$ and its standard error noted at the bottom of the table in column (3). The product-level import price index is constructed using $\hat{\sigma}$ from column (6) of Table IV according to equation (11), and the instrument is constructed using the statutory tariffs using equation (12). All regressions include sector-time fixed effects. 90% bootstrap confidence intervals are constructed from 1,000 samples. Regressions are clustered by HS-8. Significance: * 0.10, ** 0.05, *** 0.01. Sample: monthly product-level import data from 2017:1 to 2019:4.

(a) **Table V: Product Elasticity.** Product-level response used to estimate η .

TABLE VI
SECTOR ELASTICITY κ

	$\Delta \ln \left(\frac{P_{Mgt} M_{gt}}{P_{Dgt} D_{gt}} \right)$ (1)	$\Delta \ln \left(\frac{P_{Mgt}}{P_{Dgt}} \right)$ (2)	$\Delta \ln \left(\frac{P_{Mgt} M_{gt}}{P_{Dgt} D_{gt}} \right)$ (3)
$\Delta \ln Z_{Mgt}$	0.30 (0.36)	-1.59 (3.49)	
$\Delta \ln \left(\frac{P_{Mgt}}{P_{Dgt}} \right)$			-0.19 (0.49)
Sector FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes
1st-stage F			0.2
$\hat{\kappa}$ (se[$\hat{\kappa}$])			1.19 (0.49)
Bootstrap CI			[0.89, 1.71]
R^2	0.24	0.67	—
N	2,041	2,041	2,041

Notes. Table reports the sector-level import responses to import tariffs. The sample is at the sector-time level from 2017:1 to 2019:4. Column (1) reports the reduced-form regression of the imported sector's share within total sectoral expenditures imports, s_{gt} , on the product-level instrument, Z_{gt} . Column (2) reports the first stage: the regression of the product-level import price index P_{gt} on Z_{gt} . Column (3) reports the IV regression with the implied $\hat{\kappa}$ and its standard error noted at the bottom of the table in column (3). The sector import price index is constructed using $\hat{\sigma}$ from Table IV column (6) and $\hat{\eta}$ from Table V column (3) according to equation (11).

(b) **Table VI: Sector Elasticity.** Sector-level response used to estimate κ .

6.6 Table VII: Foreign Import Demand; Table VIII: Aggregate Impacts

Read:

- Table VII reports export responses to retaliatory tariffs and estimates foreign import demand facing U.S. exporters.
- Retaliatory tariffs sharply reduce U.S. export quantities and values.
- Table VIII decomposes aggregate welfare into import buyer losses, producer gains, and tariff revenue.
- The full 2018 trade war generates a large import-price cost but a much smaller net aggregate loss after revenue and producer gains.

Economic message: The export-side estimates are necessary because retaliation changes both aggregate welfare and political incidence. The aggregate table makes the distributional logic explicit: large gross losses can coexist with small net losses because transfers and producer gains offset part of the buyer loss.

Detailed interpretation: The welfare table should not be read as saying the trade war was harmless. It says that the national accounting net is small relative to the gross redistribution. Buyers of imports lose roughly \$51 billion, producers gain, the government collects revenue, and the net equivalent variation is around -\$7.2 billion.

FOREIGN IMPORT DEMAND σ^*						
	$\Delta \ln p_{\text{gr}}^X \tau_{\text{gr}}^*$ (1)	$\Delta \ln x_{\text{gr}}$ (2)	$\Delta \ln p_{\text{gr}}^X$ (3)	$\Delta \ln p_{\text{gr}}^X (1 + \tau_{\text{gr}}^*)$ (4)	$\Delta \ln p_{\text{gr}}^X$ (5)	$\Delta \ln x_{\text{gr}}$ (6)
$n(1 + \tau_{\text{gr}}^*)$	-0.99*** (0.28)	-1.00*** (0.36)	-0.04 (0.16)	0.96*** (0.16)		
αx_{gr}					0.04 (0.16)	
$n p_{\text{gr}}^X (1 + \tau_{\text{gr}}^*)$						-1.04*** (0.32)
duct $\times$ time FE	Yes	Yes	Yes	Yes	Yes	Yes
untry $\times$ time FE	Yes	Yes	Yes	Yes	Yes	Yes
untry $\times$ sector FE	Yes	Yes	Yes	Yes	Yes	Yes
stage P					7.8	38.2
otstrap CI					[-0.30, 0.26]	[0.73, 1.39]
	0.07	0.07	0.06	0.06	—	0.51
	3,306,766	2,564,731	2,564,731	2,564,731	2,564,731	2,564,731

Notes: Table reports the variety-level export responses to retaliatory tariffs. Columns (1)–(4) report reduced-form regressions of export values, quantities, before-duty unit values, duty-inclusive unit values on $\Delta \ln(1 + \tau_{\text{gr}}^*)$, the change in retaliatory export tariffs. Column (5) reports the IV regression that estimates the U.S. export supply elasticity σ^* ; the first σ^* is in column (2). Column (6) reports the IV regression that estimates the foreign import demand elasticity σ^* ; the first stage is column (4). All regressions include product-time, industry-time, and country-sector fixed effects. Standard errors are clustered by country and HS-6. 90% bootstrap confidence intervals are constructed from 1,000 samples. Significance: 10, ** 0.05, *** 0.01. Sample: monthly variety-level export data from 2017:1 to 2019:4.

(a) **Table VII: Foreign Import Demand.** Export-side responses to retaliatory tariffs.

TABLE VIII
AGGREGATE IMPACTS

	EV^M (1)	EV^X (2)	ΔR (3)	EV (4)
2018 trade war				
Change (\$ b)	-51.0 [-54.8, -47.2]	9.4 [4.1, 15.6]	34.3 [32.3, 36.1]	-7.2 [-14.4, 0.8]
Change (% GDP)	-0.27 [-0.29, -0.25]	0.05 [0.02, 0.08]	0.18 [0.17, 0.19]	-0.04 [-0.08, 0.00]
2018 U.S. tariffs and no retaliation				
Change (\$ b)	-50.9 [-52.9, -49.0]	16.6 [13.2, 20.3]	34.8 [32.8, 36.5]	0.5 [-4.0, 5.7]
Change (% GDP)	-0.27 [-0.28, -0.26]	0.09 [0.07, 0.11]	0.19 [0.18, 0.20]	0.00 [-0.02, 0.03]

Notes: Table reports the aggregate impacts in column (4) and the decomposition into EV^M , EV^X , and tariff revenue (ΔR) in columns (1)–(3). The top panel reports the effects from the 2018 trade war. The bottom panel simulates a hypothetical scenario where trade partners do not retaliate against U.S. tariffs. The first row in each panel reports the overall impacts of each term in billions of US\$. The third row scales by 2016 GDP. These numbers are computed using the model described in Section V with $\{\delta = 2.53, \eta = 1.53, \epsilon = 1.19, \phi^* = -0.00, \delta^* = 1.04\}$. Bootstrapped 90% confidence intervals based on 1,000 simulations of the estimated parameters are reported in brackets.

(b) **Table VIII: Aggregate Impacts.** Equivalent-variation decomposition into import losses, producer gains, and tariff revenue.

6.7 Figure IV: Dynamic Specification

Read:

- Figure IV plots cumulative dynamic effects for imports and exports.
- Import value and quantity losses accumulate after U.S. tariffs.
- Duty-inclusive import unit values rise, while before-duty import unit values remain much flatter.
- Export value and quantity responses to retaliation are negative but somewhat noisier.

Economic message: Figure IV confirms that the baseline regressions are not only an immediate one-month effect; the trade responses persist over the post-tariff horizon.

Detailed interpretation: The figure is useful because it shows the timing of adjustment. Quantity responses build after tariffs, while price pass-through appears quickly. This timing supports the interpretation that the policy shock changed trade costs rather than merely capturing unrelated product trends.

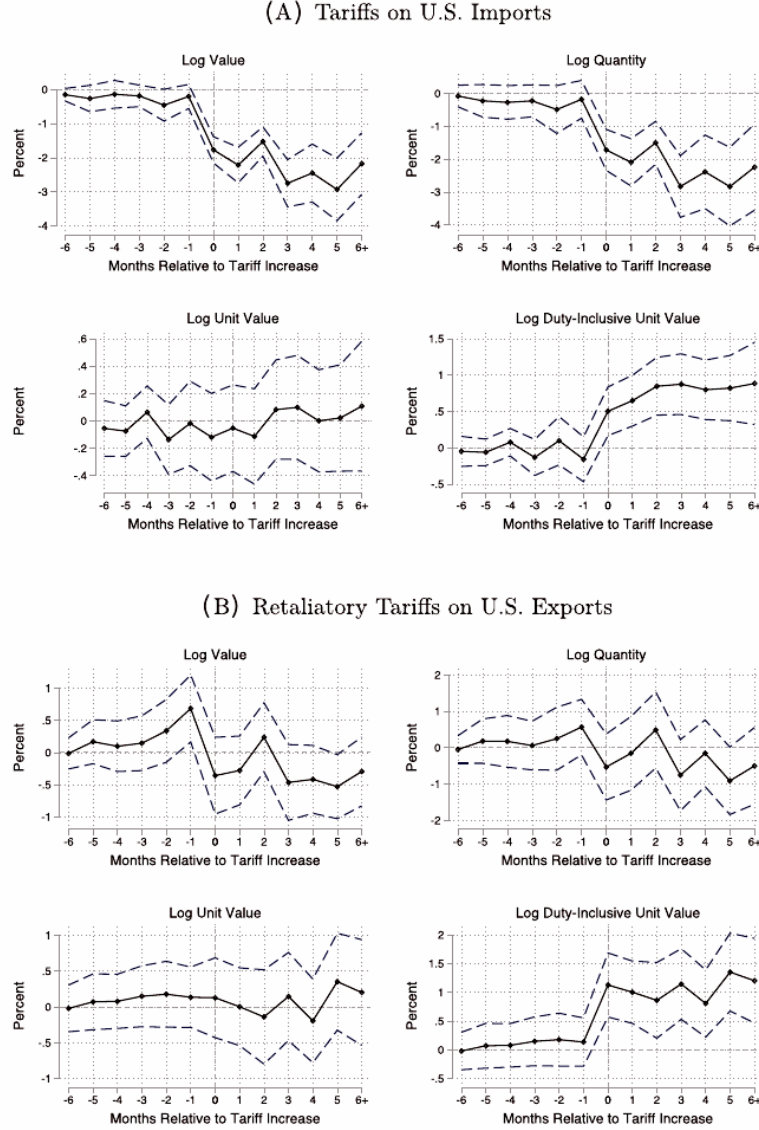


FIGURE IV

Dynamic Specification

Figure IV: Dynamic Specification. Cumulative event-study coefficients for import and export outcomes.

6.8 Figure V: Regional Variation in U.S. and Retaliatory Tariffs; Figure VI: Model Simulation of Real Wage Impacts

Read:

- Figure V maps county-level exposure to U.S. import tariffs and retaliatory export tariffs.
- Import protection is concentrated in parts of the industrial Midwest and Northeast.
- Retaliation is more concentrated in rural and agricultural regions, especially the Plains and Mountain West.
- Figure VI maps model-implied real wage losses with and without retaliation.
- Every county loses in real tradeable wage terms, but losses are much larger when retaliation is included.

Economic message: The maps show why the aggregate effect is not the whole story. The trade war created geographically uneven burdens, and retaliation altered which regions lost most.

Detailed interpretation: Figure V is an exposure map; Figure VI is an incidence map. Exposure does not translate mechanically into welfare because sectoral linkages, consumption price changes, and retaliatory tariffs all matter. The contrast between the two figures shows why a model is needed for regional incidence.

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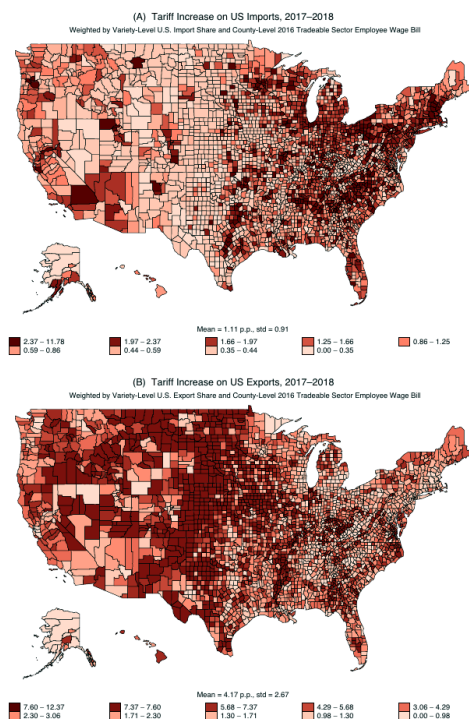


FIGURE V

Regional Variation in U.S. and Retaliatory Tariffs

(a) **Figure V: Regional Variation in U.S. and Retaliatory Tariffs.** County exposure to import and export tariff increases.

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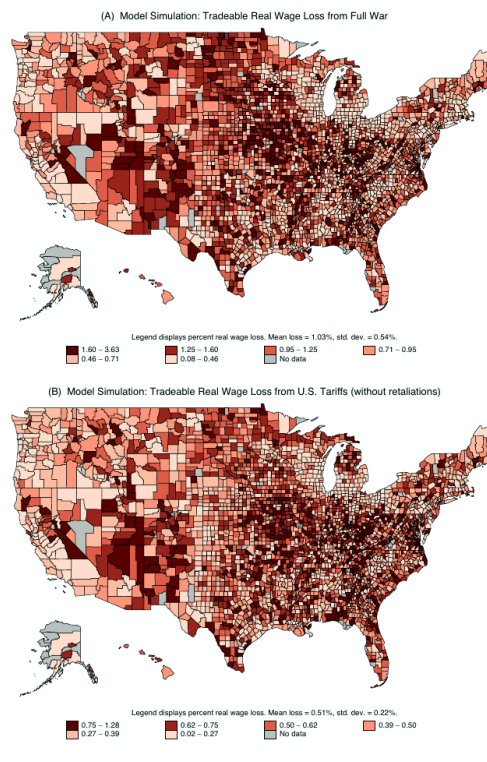


FIGURE VI

Model Simulation of Real Wage Impacts from U.S. and Retaliatory Tariffs

Figure shows county-level mean tradeable wage losses as simulated from the model. Panel A shows losses accounting for both import and retaliatory tariffs.

(b) **Figure VI: Model Simulation of Real Wage Impacts from U.S. and Retaliatory Tariffs.** County-level real wage losses with and without retaliation.

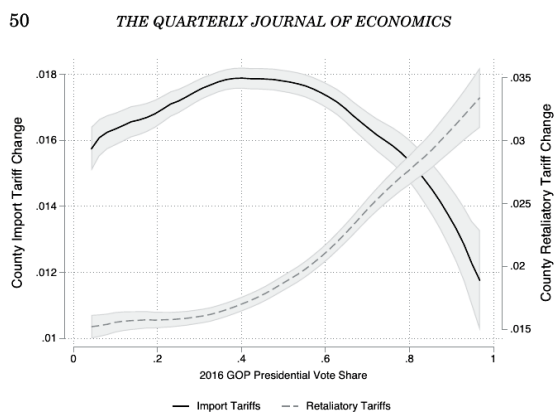
6.9 Figure VII: Tariff Changes versus 2016 Republican Vote Share; Figure VIII: Model-Simulated Tradeable Real Wage Impact versus 2016 Republican Vote Share

Read:

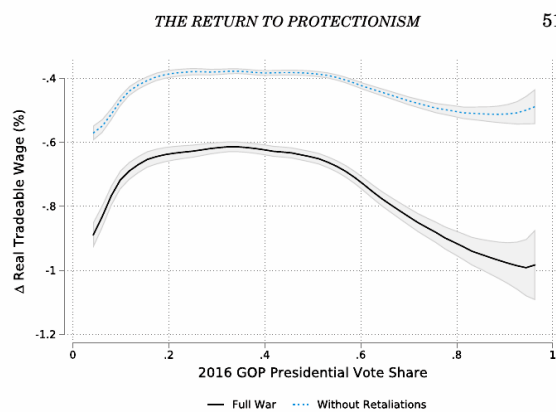
- Figure VII shows an inverted-U relation between U.S. tariff protection and Republican vote share.
- Counties with roughly competitive vote shares receive more import-tariff protection.
- Retaliatory tariffs rise more strongly with Republican vote share.
- Figure VIII shows that heavily Republican counties bear the largest real wage losses in the full trade war.
- Without retaliation, the regional political pattern would be much less punitive to Republican counties.

Economic message: The political-economy message is sharp: U.S. protection appears targeted toward electorally competitive areas, but foreign retaliation undermines the intended regional protection by hitting Republican-leaning export sectors.

Detailed interpretation: Figure VII is about policy targeting; Figure VIII is about realized incidence. The distinction matters. A tariff can be designed to help politically important regions, but once trading partners retaliate, the ex post burden can fall on the incumbent's own political base.



(a) **Figure VII: Tariff Changes versus 2016 Republican Vote Share.** County tariff exposure by GOP vote share.



(b) **Figure VIII: Model-Simulated Tradeable Real Wage Impact versus 2016 Republican Vote Share.** Tradeable real wage losses by GOP vote share.

7 Short summary

- The paper studies the 2018 U.S. trade war using product-country-month variation in U.S. tariffs and foreign retaliation.
- U.S. import tariffs reduced import quantities and values.
- Before-duty import prices did not fall, implying that foreign exporters did not absorb much of the tariff.
- Duty-inclusive import prices increased, so U.S. consumers and importing firms bore the short-run tariff cost.
- Retaliatory tariffs reduced U.S. export quantities and values.
- The paper estimates product, sector, and export demand elasticities to discipline a quantitative model.
- Import buyer losses are large, around \$51 billion annually.
- Tariff revenue and producer gains offset part of these losses, leaving a small aggregate real-income loss of around \$7.2 billion.
- Regional effects are large because different counties specialize in different sectors.
- U.S. tariff protection was concentrated in electorally competitive counties.
- Retaliatory tariffs were concentrated in sectors located in heavily Republican counties.
- Model simulations imply that heavily Republican counties bore the largest tradeable real wage losses.
- The main lesson is that tariffs are redistributive: aggregate losses may look small while incidence across firms, consumers, workers, and regions is large.

8 Conclusion

8.1 Core conclusion

Fajgelbaum, Goldberg, Kennedy, and Khandelwal show that the 2018 trade war imposed large short-run costs on U.S. buyers of imported goods and substantially reduced trade flows. The tariffs were passed through almost completely to duty-inclusive import prices, so the incidence fell primarily on U.S. consumers and firms using imported inputs rather than on foreign exporters.

8.2 Economic intuition

The core intuition is simple. A tariff raises the wedge between what U.S. buyers pay and what foreign sellers receive. If foreign export prices do not fall, then the tariff raises the U.S. buyer price one-for-one. Buyers respond by reducing demand, but not enough to undo the price increase. Producer gains and tariff revenue partially offset buyer losses at the national level, but this accounting hides redistribution across regions and sectors.

8.3 Incremental contribution revisited

The paper’s incremental contribution is to combine three layers in one empirical framework. First, it measures tariff incidence directly with high-frequency product-country trade data. Second, it estimates the elasticities needed to discipline a quantitative model. Third, it maps the tariff war into regional and political incidence. This combination allows the paper to move from “tariffs raised prices” to “which regions, sectors, and political constituencies bore the consequences.”

8.4 Policy implications

The findings imply that unilateral tariffs can be expensive even when the national net welfare number looks modest. The direct losses to buyers are large, and foreign retaliation can reverse the intended political incidence of protection. The paper also suggests that policymakers should evaluate trade policy as a distributional intervention, not only as an aggregate-efficiency instrument.

8.5 Limitations and caveats

The authors are explicit that their analysis focuses on the short run. It does not measure retail prices, long-run supply-chain reorganization, trade-policy uncertainty, investment responses, or foreign wage adjustments. The aggregate welfare loss could therefore be larger or smaller under longer horizons or richer market structure assumptions. The cleanest evidence is on short-run import and export responses and short-run tariff pass-through.

8.6 Takeaway

The broad takeaway is that the return to protectionism produced large redistribution with limited aggregate gains. Tariffs shifted resources toward protected producers and the government, but at substantial cost to buyers and with major political-geographic side effects once retaliation occurred.